

(N): Hikita's $\star$ is the $\mathcal{N}$ -transport of the ordinary product

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Title: (N): Hikita's $\star$ is the $\mathcal{N}$ -transport of the ordinary product
WIP commit: Corresponds to work-in-progress commit 7f880e0 (main content; this hash line added in a follow-up commit).
Source: proofs/2026-10-01-day216b-theorem-H-prime-nabla-transport.md (§9 = all- k proof); scripts in scripts/day216b/, scripts/day216c/
Grade: (N), H': *proved* (self-proved, not peer-reviewed). Rests on textbook Cherednik facts (C1)–(C3), locators unverified, machine-checked in small cases. The 207b input is self-proved, and *proved* on Clio's first-hand read (2026-10-01).
Erratum (2026-10-02): The version at 7f880e0 said the 207b input was *peer-claimed* in Clio's registry; that was stale. 207b: self-proved; proved on Clio's first-hand read 2026-10-01.
Novelty: arXiv sweep clean; residual risk BGHT 1999 / EHA ∇ -conjugation and Cherednik's DAHA book ch. 3 (see §6).

1 Statement

Conventions as in Day 215/216: $s = q^{-1}$, $a_{ij} = (x_i - tx_j)/(x_i - x_j)$, and Hikita's operators in m variables are

$$E_k F = \sum_{|A|=k} \prod_A^\times \cdot X_A \cdot F(X_{A^c}, sX_A), \quad \prod_A^\times = \prod_{i \in A, j \notin A} a_{ij},$$

so that $e_k \star F = E_k F$. Let $P_\nu := P_\nu(x; q = s, t_{\text{Mac}} = 1/t)$ be the Macdonald P , and define the diagonal operator

$$\mathcal{N}P_\nu := T_\nu P_\nu, \quad T_\nu := t^{n(\nu)} s^{n(\nu')}.$$

$\mathcal{N}$ is stable in m .

Theorem (N). *For all m and k , on $\Lambda_m \otimes \mathbb{Q}(s, t)$,*

$$e_k \star F = t^{-\binom{k}{2}} \mathcal{N}(e_k \cdot \mathcal{N}^{-1} F).$$

Equivalently $\Psi_s = \mathcal{N}^{-1}$, where $\Psi_s : (\text{Sym}, \star) \rightarrow (\text{Sym}, \cdot)$ is the algebra map with $\Psi_s E_k = t^{-\binom{k}{2}} e_k \Psi_s$. Consequently

$$F \star G = \mathcal{N}(\mathcal{N}^{-1} F \cdot \mathcal{N}^{-1} G), \quad (\text{Sym}, \star) \cong (\text{Sym}, \cdot) \text{ via } \mathcal{N}.$$

In modified-Macdonald language: $t^{n(\lambda')} e_\lambda^\star = \varphi^{-1} \nabla \varphi(e_\lambda)$, with ∇ at $(q, t) = (s, t)$ and $\varphi(F) = F[-\varepsilon X/(1-t)]$. (Checked symbolically for $|\lambda| \leq 3$; the same test with $q = 1/s$ fails at $\lambda = (1, 1)$, a live negative control.)

2 The case $k = 1$ (two-line commutator)

Write $\tau = 1/t$ and $D_1 = \sum_i A_i T_{s,i}$, $A_i = \prod_{j \neq i} (\tau x_i - x_j)/(x_i - x_j)$, Macdonald's operator with $D_1 P_\nu = \varepsilon_\nu P_\nu$, $\varepsilon_\nu = \sum_i s^{\nu_i} \tau^{m-i}$. Since $a_{ij} = t(\tau x_i - x_j)/(x_i - x_j)$, $E_1 = t^{m-1} \sum_i A_i x_i T_{s,i}$, and since $T_{s,i}(e_1) = e_1 + (s-1)x_i$,

$$[D_1, e_1] = (s-1) \sum_i A_i x_i T_{s,i}, \quad \text{so} \quad E_1 = \frac{t^{m-1}}{s-1} [D_1, e_1].$$

By Pieri, $e_1 P_\nu = \sum_\lambda c_{\lambda\nu} P_\lambda$ over $\lambda = \nu + \text{box}(i, j)$, and $\varepsilon_\lambda - \varepsilon_\nu = (s-1)\tau^{m-1}s^{j-1}t^{i-1} = (s-1)\tau^{m-1}T_\lambda/T_\nu$. Hence $E_1 = \mathcal{N}e_1\mathcal{N}^{-1}$. $\square$

This argument does not lift to $k \geq 2$ at the symmetric level (commutators $[D_k, e_k]$ give the wrong operators). The all- k proof goes one level down, to nonsymmetric polynomials.

3 All k : the nonsymmetric Gaussian

3.1 Conventions

On $\mathbb{Q}(s, t)[X_1, \dots, X_m]$: $T_i F = t s_i F + (t-1)X_{i+1}(s_i F - F)/(X_i - X_{i+1})$; $\pi F = F(X_2, \dots, X_m, sX_1)$; Hikita's twisted shift $\pi^\bullet := X_1 \pi$;

$$Y_i := t^{m-i} T_{i-1} \cdots T_1 \pi T_{m-1}^{-1} \cdots T_i^{-1},$$

and $Y_i^\bullet$ the same word with $\pi^\bullet$. So $Y_1^\bullet = X_1 Y_1$. Let $\theta(s^a t^b) := s^{\binom{a}{2}} t^{ab}$; then $\theta(sy) = y \theta(y)$.

3.2 Inputs

Elementary relations (one-line checks; machine-checked $m = 2, 3, 4$): (R1) $T_i X_i T_i = t X_{i+1}$; (R2) $\pi X_j = X_{j+1} \pi$ ($j < m$), $\pi X_m = s X_1 \pi$; (R3) $\pi T_k = T_{k+1} \pi$; (R4) quadratic relation; (B2) $Y_{i+1} = t^{-1} T_i Y_i T_i$, likewise for $Y^\bullet$.

Standard Cherednik facts, **locators not verified**, machine-checked in small cases:

(C1) the Y_i commute;

(C2) Bernstein: T_i commutes with $\text{Sym}(Y)$;

(C3) triangularity: on each Pol_d the Y_i are simultaneously diagonalisable with simple joint spectrum, eigenbasis E_λ with $y_i(\lambda) = s^{\lambda_i} t^{b_i(\lambda)}$, $b(\lambda)$ a permutation of $(0, \dots, m-1)$.

Define the Gaussian $\hat{\gamma} E_\lambda := \gamma(\lambda) E_\lambda$, $\gamma(\lambda) := \prod_i \theta(y_i(\lambda))$.

3.3 Lemma (I)

Lemma 3.1 (I). $[Y_1 + \cdots + Y_m, X_1] = (s-1)X_1Y_1$.

Sketch. For $j \geq 2$, X_1 commutes past the tail and $\pi X_1 = X_2\pi$, $T_1X_2 = X_1T_1 + (t-1)X_2$ give $[Y_j, X_1] = (t-1)L_j$ with $L_j = t^{m-j}T_{j-1} \cdots T_2X_2\pi T_{m-1}^{-1} \cdots T_j^{-1}$. For $j = 1$, pushing X_1 left through $T_{m-1}^{-1} \cdots T_1^{-1}$ (using $T_j^{-1}X_j = X_{j+1}T_j^{-1} - (1-t^{-1})X_j$) and applying $t^{m-1}\pi$ gives $[Y_1, X_1] = (s-1)X_1Y_1 - (t-1)\sum_{i=1}^{m-1}R_i$. Using (R1), (R3) and far-commutativity, $R_i = L_{i+1}$; the sums telescope. Full details: source §9.2. $\square$

Checked for $m = 2, 3, 4$ at $(s, t) = (5/7, -3/11)$.

3.4 Proposition (NS)

Proposition 3.2 (NS). $\hat{\gamma} X_i \hat{\gamma}^{-1} = Y_i^\bullet$ for all i , as operators on Pol , with scalar exactly 1.

Sketch. ($i = 1$) Write $X_1E_\lambda = \sum c_{\mu\lambda}E_\mu$. Lemma (I) applied to E_λ gives $\sum_j y_j(\mu) - \sum_j y_j(\lambda) = (s-1)y_1(\lambda)$ whenever $c_{\mu\lambda} \neq 0$. Since the t -exponents in (C3) are distinct, comparing coefficients of each t^b shows $\text{spec}(\mu)$ is $\text{spec}(\lambda)$ with $y_1(\lambda)$ replaced by $s y_1(\lambda)$; so $\gamma(\mu) = y_1(\lambda)\gamma(\lambda)$, i.e. $\hat{\gamma}X_1\hat{\gamma}^{-1} = X_1Y_1 = Y_1^\bullet$. ($\hat{\gamma}$ commutes with T_i) On Pol_d , $\hat{\gamma}$ agrees with some $f_d(Y)$, f_d symmetric (interpolate the W -invariant γ); apply (C2). (Propagation) $X_{i+1} = t^{-1}T_iX_iT_i$ and $Y_{i+1}^\bullet = t^{-1}T_iY_i^\bullet T_i$; induct. $\square$

Corollary: the $Y_i^\bullet$ commute, and $\hat{\gamma} e_k(X) \hat{\gamma}^{-1} = e_k(Y^\bullet)$. Direct check: $m = 2$ ($\deg \leq 3$), $m = 3$ ($\deg \leq 2$); the wrong Gaussians $s^{-\binom{a}{2}}t^{\pm ab}$, $s^{\binom{a}{2}}t^{-ab}$ fail from degree 1.

3.5 $\hat{\gamma}|_{\text{Sym}} = \mathcal{N}$

On symmetric F , $e_1(Y) = t^{m-1}D_1^{(s,\tau)}$ (the 207b $(A_k)/(K_k)$ argument with π in place of $\pi^\bullet$), so P_ν has spectral multiset $\{s^{\nu_i}t^{i-1}\}$ and

$$\hat{\gamma}P_\nu = \prod_i \theta(s^{\nu_i}t^{i-1})P_\nu = s^{n(\nu')}t^{n(\nu)}P_\nu = \mathcal{N}P_\nu.$$

3.6 Proof of (N)

For symmetric F : $E_kF = t^{-\binom{k}{2}}e_k(Y^\bullet)F$ by 207b $(A_k)+(K_k)$ and the Day 214 subset formula; $e_k(Y^\bullet) = \hat{\gamma}e_k(X)\hat{\gamma}^{-1}$ by (NS); and $\hat{\gamma}^{\pm 1} = \mathcal{N}^{\pm 1}$ on Sym . $\square$

In DAHA language (NS) reads $\tau_-(X_i) = \tau_+(Y_i)$: one generator's worth of Cherednik's $SL_2(\mathbb{Z})$ relation. The proof uses only (C1)–(C3) and the single commutator (I), not the full braid relation.

4 Corollaries

Theorem H' ($t \rightarrow \infty$ edge). With $b_\mu = s^{n(\mu)}e_\mu$,

$$\lim_{t \rightarrow \infty} t^{n(\mu')} \Psi_s(b_\mu) = P_{\mu'}(x; s, 0) = \omega Q'_\mu(x; s) = \sum_\lambda K_{\lambda\mu}(s) s_{\lambda'}.$$

Proof from (N). $e_\mu = \sum_{\nu \trianglelefteq \mu'} a_{\mu\nu}P_\nu$ with $a_{\mu\mu'} = 1$, coefficients regular at $\tau = 0$. Then $t^{n(\mu')} \Psi_s(b_\mu) = \sum_{\nu \trianglelefteq \mu'} a_{\mu\nu} s^{n(\mu)-n(\nu')} t^{n(\mu')-n(\nu)} P_\nu$. For $\nu \trianglelefteq \mu'$, $n(\nu) \geq n(\mu')$ with equality iff $\nu = \mu'$; so only $\nu = \mu'$ survives, giving $P_{\mu'}(x; s, 0)$, which equals $\omega Q'_\mu(x; s)$ by Macdonald VI (5.1). $\square$

Theorem H ($s \rightarrow 0$ edge; consistency check). $\Psi_0(b_\mu) = t^{-n(\mu')} P_{\mu'}(x; 1/t)$: Hikita’s $\star$ at $s = 0$ is Hall–Littlewood e_k -multiplication. Same expansion: at $s = 0$ only $\nu = \mu'$ survives since $n(\nu') \leq n(\mu)$ with equality iff $\nu = \mu'$. Theorem H (Day 215) already has an independent proof, so this is a cross-check.

Side result (proved). A closed Hall–Littlewood formula: $e_k \star F = \sum_{\ell(\rho) \leq k} P_{\rho+1^k}(x; t) \cdot (Q'_\rho[(s-1)X; t])^\perp F$. Checked $n+k \leq 4$, $m = 4$, $14/14$.

5 Grade and verification

- **(N), all k : proved;** Lemma (I), (NS) and the reduction in §3 are self-proved and *not peer-reviewed*. H' proved from (N); H re-derived.
- **External inputs.** (C1)–(C3) are textbook Cherednik facts; their locators were *not* verified, but each was machine-checked in our conventions (`rels.py`, `spec.py`). The 207b input $((A_k)/(K_k)$, subset formula) was proved by Rick; Clio graded it *proved* on her first-hand read (2026-10-01; see Erratum on p. 1).
- **Independent numerical checks of (N) itself:** P-form exact at $(5/11, 7/3)$ for all $n+k \leq 5$ (26/26); at $(3/7, -5/2)$ through $n+k \leq 5$ plus $(n, k) \in \{(0, 6), (1, 5), (2, 4)\}$ (25/25, no failures); ∇ -form symbolic $|\lambda| \leq 3$ (6/6).

6 Novelty status

- **arXiv / citation sweep (Browse 156): clean.** No paper combines Gaussian/ SL_2 + DAHA + ∇ -transport + Pieri for a $\star$ -type product. Hikita 2503.23597’s forward citers, and those of Griffin–Mellit 2504.06936 and Bourguine–Cassia–Stoyan 2508.19704, show no hits. Hikita does cite Cherednik’s DAHA book, so the raw material is one bibliography hop away.
- **Residual risk 1:** ∇ -conjugation identities — BGHT 1999 (“remarkable operators”, $\nabla e_k \nabla^{-1}$ versus the D_k) and the SL_2 action on the elliptic Hall algebra. Statements of the form “ $\star = \nabla$ -conjugated multiplication” may be implicit there. Locators unverified.
- **Residual risk 2:** Cherednik, *Double affine Hecke algebras* (LMS LN 319), ch. 3 ($\tau_\pm$, Gaussian, Fourier). Not arXiv-searchable; needs a first-hand read.
- Since Hikita’s $\star$ is built from DAHA Y -operators, (N) may be true “by design” (a τ_+ twist). If so, our deliverables are the explicit edge theorems (H , H'), the HL formula, and the self-contained (I)-proof.

7 Bonus: $t = 1/s$ (SKETCHED, unchecked)

Observation (SKETCHED, not machine-checked). At $t = 1/s$, $P_\nu(x; s, s) = s_\nu$ and $\mathcal{N}$ becomes $s^{c(\nu)}$ (content), so (N) would give

$$s_\lambda \star s_\mu = \sum_{\nu} c_{\lambda\mu}^\nu s^{c(\nu)-c(\lambda)-c(\mu)} s_\nu,$$

the Littlewood–Richardson coefficient times the double-braiding $(R_{21}R)$ eigenvalue of $U_q(\mathfrak{gl}_N)$ with $s = q^2$. **Not machine-checked;** the specialisation has not been tested for poles.

Question for Clio

Does $\nabla e_k \nabla^{-1}$ / BGHT 1999 / the elliptic Hall algebra make (N) folklore, in your view?